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Forceps with linear trigger mechanism

Abstract

A forceps includes first and second shafts each having a jaw disposed at a distal end thereof configured to rotate about a pivot to move the jaws between open and closed positions, the shafts defining a longitudinal axis therebetween. A knife deployment mechanism including a rack and pinion mechanism or a series of links is disposed within the first shaft and includes a trigger moveable along the longitudinal axis to deploy a knife between the jaws. A knife lockout is configured to move upon approximation of the first and second shafts between an engaged position preventing deployment of the knife and a disengaged position allowing deployment of the knife. A knife kickout mechanism is disposed within the first shaft in opposition to the second shaft and is configured to force the knife forward upon movement of the first and second shafts from an approximated position to a more open position.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of and priority to U.S. provisional application Ser. No. 62/990,277, filed on Mar. 16, 2020, the entire content of which is incorporated by reference herein.

BACKGROUND

Technical Field

(1) The present disclosure relates to electrosurgical instruments and, more particularly, to electrosurgical forceps for grasping, treating, and/or dividing tissue.

Background of Related Art

(2) A surgical forceps is a plier-like instrument which relies on mechanical action between its jaws to grasp tissue. Electrosurgical forceps utilize both mechanical clamping action and electrical energy to treat tissue, e.g., coagulate, cauterize, and/or seal tissue.

(3) Typically, once tissue is treated, the surgeon has to accurately sever the treated tissue. Accordingly, many electrosurgical forceps have been designed which incorporate a knife configured to effectively sever tissue after treating the tissue.

SUMMARY

(4) As used herein, the term “distal” refers to the portion that is being described which is further from a surgeon, while the term “proximal” refers to the portion that is being described which is closer to a surgeon. Further, to the extent consistent, any of the aspects described herein may be used in conjunction with any or all of the other aspects described herein.

(5) An electrosurgical forceps provided in accordance with aspects of the present disclosure

includes first and second shaft members each having a jaw member disposed at a distal end thereof, the first and second shaft members configured to rotate about a pivot to move the jaw members between an open position and a closed position, the first and second shaft members defining a longitudinal axis therebetween. A knife deployment mechanism is disposed within the first shaft member and includes a trigger moveable along the longitudinal axis to deploy a knife (which may also be deployable along the longitudinal axis) operably coupled thereto between a retracted position relative to the jaw members and an extended position between the jaw members. The knife deployment mechanism includes first and second rack members operably coupled to one another by a gear disposed therebetween. The trigger is operably connected to the first rack member and the knife is operably coupled to the second rack member such that movement of the trigger moves the knife in an opposite direction relative thereto. A knife lockout is configured to move upon approximation of the first and second shaft members between an engaged position preventing deployment of the knife and a disengaged position allowing deployment of the knife. The knife lockout includes a flange operably connected to the first shaft member and depending therefrom in opposition to the second shaft member such that approximation of the first and second shaft members forces the flange against the second shaft member to disengage the knife lockout to allow actuation of the knife.

(6) In aspects according to the present disclosure, the first shaft member includes a trigger slot defined therein, the trigger is configured to travel between a distal-most position wherein the trigger slot is exposed and a more proximal position wherein the trigger covers the trigger slot to reduce the chances of a user's finger being pinched within the trigger slot. In other aspects according to the present disclosure, the forceps further includes a switch assembly disposed on one of the first or second shaft members which is configured to be engaged by the other of the first or second shaft members when the jaw members are approximated to move the switch assembly between a deactivated position and an activated position to control delivery of electrosurgical energy to the jaw members.

(7) In aspects according to the present disclosure, a knife return spring is operably coupled to the knife deployment mechanism and is configured to bias the knife toward the retracted position. In other aspects according to the present disclosure, the knife return spring is operable coupled to one or both of the first or second rack members. In yet other aspects according to the present disclosure, the knife lockout includes a slot defined in the flange configured to operably engage a lock pin disposed in the knife deployment mechanism to prevent movement of the knife when engaged.

(8) In aspects according to the present disclosure, upon approximation of the first and second shaft members, the flange is configured to dislodge the slot from engagement with the lock pin of the knife deployment mechanism to allow selective actuation of the knife. In other aspects according to the present disclosure, the flange is connected to the first shaft member by a flange pin. In yet other aspects according to the present disclosure, the flange is fixed at a distal end thereof by the flange pin and, upon approximation of the first and second shaft members, the flange is configured to cantilever or flex about the flange pin to dislodge the lock pin from the slot defined therein. In yet other aspects according to the present disclosure, upon opening of the first and second shaft members relative to one another the bias of the flange reseats the lock pin within the slot.

(9) In aspects according to the present disclosure, the knife deployment mechanism includes an elongated slot defined therein to allow reciprocation of the lock pin therein. In other aspects according to the present disclosure, the flange includes a ramp to facilitate reseating the lock pin within the slot upon return of the knife deployment mechanism.

(10) In yet other aspects according to the present disclosure, the knife lockout includes a boss disposed on the flange configured to operably engage one of a plurality of slots defined between a plurality of gears in the first rack to prevent movement of the knife when engaged. In still other aspects according to the present disclosure, upon approximation of the first and second shaft members, the boss on the flange is configured to dislodge from the one of a plurality of slots to

allow selective actuation of the knife. In aspects according to the present disclosure, the flange is fixed at a proximal end thereof by the pivot and, upon approximation of the first and second shaft members, the flange is configured to rotate about the pivot to dislodge the boss from gear.

(11) In aspects according to the present disclosure, a knife kickout mechanism is configured to force the knife forward upon movement of the first and second shaft members from an approximated position to a more open position. In other aspects according to the present disclosure, the knife kickout mechanism includes a flange depending from the knife deployment mechanism in oppositional alignment with the second shaft member wherein, upon approximation of the first and second shaft members and actuation of the knife deployment mechanism in a first direction, the knife kickout rides within a slot defined within the second shaft member to abuttingly engage a ramp defined in the slot and wherein, upon opening of the first and second shaft members relative to one another, the ramp forces the knife kickout mechanism in an opposite direction to facilitate return of the knife deployment mechanism to an unactuated position.

(12) An electrosurgical forceps provided in accordance with additional aspects of the present disclosure includes first and second shaft members each having a jaw member disposed at a distal end thereof, the first and second shaft members configured to rotate about a pivot to move the jaw members between an open position and a closed position, the first and second shaft members defining a longitudinal axis therebetween. A knife deployment mechanism is disposed within the first shaft member and includes a trigger moveable along the longitudinal axis to deploy a knife operably coupled thereto between a retracted position relative to the jaw members and an extended position between the jaw members, the knife deployment mechanism including first and second rack members operably coupled to one another by a gear disposed therebetween, the trigger operably connected to the first rack member and the knife operably coupled to the second rack member such that movement of the trigger moves the knife in an opposite direction relative thereto. A knife kickout mechanism is configured to force the knife forward upon movement of the first and second shaft members from an approximated position to a more open position, the knife kickout mechanism including a flange depending from the knife deployment mechanism in oppositional alignment with the second shaft member wherein, upon approximation of the first and second shaft members and actuation of the knife deployment mechanism in a first direction, the knife kickout rides within a slot defined within the second shaft member to abuttingly engage a ramp defined in the slot and wherein, upon opening of the first and second shaft members relative to one another, the ramp forces the knife kickout mechanism in an opposite direction to facilitate return of the knife deployment mechanism to an unactuated position.

(13) In aspects according to the present disclosure, a switch assembly is disposed on one of the first or second shaft members and is configured to be engaged by the other of the first or second shaft members when the jaw members are approximated to move the switch assembly between a deactivated position and an activated position to control delivery of electrosurgical energy to the jaw members. In other aspects according to the present disclosure, a knife return spring is operably coupled to the knife deployment mechanism and configured to bias the knife toward the retracted position.

(14) An electrosurgical forceps provided in accordance with additional aspects of the present disclosure includes first and second shaft members each having a jaw member disposed at a distal end thereof, the first and second shaft members configured to rotate about a pivot to move the jaw members between an open position and a closed position, the first and second shaft members defining a longitudinal axis therebetween. A knife deployment mechanism is disposed within the first shaft member and includes a trigger moveable along the longitudinal axis to deploy a knife operably coupled thereto between a retracted position relative to the jaw members and an extended position between the jaw members, the knife deployment mechanism including a series of links operably coupled to one another and to a knife carrier configured to translate the knife. A knife lockout is configured to move upon approximation of the first and second shaft members between

an engaged position preventing deployment of the knife and a disengaged position allowing deployment of the knife, the knife lockout including a flange operably connected to the first shaft member and depending therefrom in opposition to the second shaft member such that approximation of the first and second shaft members forces the flange against the second shaft member to disengage the knife lockout to allow actuation of the knife.

(15) In aspects according to the present disclosure, the first shaft member includes a trigger slot defined therein, the trigger is configured to travel between a distal-most position wherein the trigger slot is exposed and a more proximal position wherein the trigger covers the trigger slot to reduce the chances of a user's finger being pinched within the trigger slot. In other aspects according to the present disclosure, a switch assembly is disposed on one of the first or second shaft members and is configured to be engaged by the other of the first or second shaft members when the jaw members are approximated to move the switch assembly between a deactivated position and an activated position to control delivery of electrosurgical energy to the jaw members.

(16) In aspects according to the present disclosure, a knife return spring is operably coupled to the knife deployment mechanism and configured to bias the knife toward the retracted position. In other aspects according to the present disclosure, the knife return spring is operable coupled to at least one of the series of links.

(17) In aspects according to the present disclosure, the knife lockout includes a slot defined in the flange configured to operably engage a lock pin disposed in the knife deployment mechanism to prevent movement of the knife when engaged. In other aspects according to the present disclosure, upon approximation of the first and second shaft members, the flange is configured to dislodge the slot from engagement with the lock pin of the knife deployment mechanism to allow selective actuation of the knife. In yet other aspects according to the present disclosure, the flange is connected to the first shaft member by a sleeve.

(18) In aspects according to the present disclosure, the flange includes an elongated shaft fixed at a distal end thereof by the sleeve disposed within the first shaft member and, upon approximation of the first and second shaft members, the elongated shaft of the flange is configured to cantilever or flex at the sleeve to dislodge the lock pin from the slot defined therein. In other aspects according to the present disclosure, upon opening of the first and second shaft members relative to one another the bias of the elongated shaft reseats the lock pin within the slot.

(19) In aspects according to the present disclosure, the knife deployment mechanism includes an elongated slot defined therein to allow reciprocation of the lock pin therein. In other aspects according to the present disclosure, the flange includes a ramp to facilitate reseating the lock pin within the slot of the flange upon return of the knife deployment mechanism.

(20) In aspects according to the present disclosure, a knife kickout mechanism is configured to force the knife forward upon movement of the first and second shaft members from an approximated position to a more open position. In other aspects according to the present disclosure, the knife kickout mechanism includes a flange depending from the knife deployment mechanism in oppositional alignment with the second shaft member wherein, upon approximation of the first and second shaft members and actuation of the knife deployment mechanism in a first direction, the knife kickout rides within a slot defined within the second shaft member to abuttingly engage a ramp defined in the slot and wherein, upon opening of the first and second shaft members relative to one another, the ramp forces the knife kickout mechanism in an opposite direction to facilitate return of the knife deployment mechanism to an unactuated position.

(21) An electrosurgical forceps provided in accordance with additional aspects of the present disclosure includes first and second shaft members each having a jaw member disposed at a distal end thereof, the first and second shaft members configured to rotate about a pivot to move the jaw members between an open position and a closed position, the first and second shaft members defining a longitudinal axis therebetween. A knife deployment mechanism is disposed within the first shaft member and includes a trigger moveable along the longitudinal axis to deploy a knife

operably coupled thereto between a retracted position relative to the jaw members and an extended position between the jaw members, the knife deployment mechanism including a series of links operably coupled to one another and to a knife carrier configured to translate the knife. A knife kickout mechanism is configured to force the knife forward upon movement of the first and second shaft members from an approximated position to a more open position, the knife kickout mechanism including a flange depending from the knife deployment mechanism in oppositional alignment with the second shaft member wherein, upon approximation of the first and second shaft members and actuation of the knife deployment mechanism in a first direction, the knife kickout rides within a slot defined within the second shaft member to abuttingly engage a ramp defined in the slot and wherein, upon opening of the first and second shaft members relative to one another, the ramp forces the knife kickout mechanism in an opposite direction to facilitate return of the knife deployment mechanism to an unactuated position.

(22) In aspects according to the present disclosure, a switch assembly disposed on one of the first or second shaft members and configured to be engaged by the other of the first or second shaft members when the jaw members are approximated to move the switch assembly between a deactivated position and an activated position to control delivery of electrosurgical energy to the jaw members.

(23) In aspects according to the present disclosure, a knife return spring operably coupled to the knife deployment mechanism and configured to bias the knife toward the retracted position

(24) An electrosurgical forceps provided in accordance with additional aspects of the present disclosure includes first and second shaft members each having a jaw member disposed at a distal end thereof, the first and second shaft members configured to rotate about a pivot to move the jaw members between an open position and a closed position, the first and second shaft members defining a longitudinal axis therebetween. A knife deployment mechanism is disposed within the first shaft member and includes a trigger moveable along the longitudinal axis to deploy a knife operably coupled thereto between a retracted position relative to the jaw members and an extended position between the jaw members. A knife lockout is configured to move upon approximation of the first and second shaft members between an engaged position preventing deployment of the knife and a disengaged position allowing deployment of the knife, the knife lockout including a flange operably connected to the first shaft member and depending therefrom in opposition to the second shaft member such that approximation of the first and second shaft members forces the flange against the second shaft member to disengage the knife lockout to allow actuation of the knife. A knife kickout mechanism is configured to force the knife forward upon movement of the first and second shaft members from an approximated position to a more open position, the knife kickout mechanism including a flange depending from the knife deployment mechanism in oppositional alignment with the second shaft member wherein, upon approximation of the first and second shaft members and actuation of the knife deployment mechanism in a first direction, the knife kickout rides within a slot defined within the second shaft member to abuttingly engage a ramp defined in the slot and wherein, upon opening of the first and second shaft members relative to one another, the ramp forces the knife kickout mechanism in an opposite direction to facilitate return of the knife deployment mechanism to an unactuated position.

(25) In aspects according to the present disclosure, the first shaft member includes a trigger slot defined therein, the trigger is configured to travel between a distal-most position wherein the trigger slot is exposed and a more proximal position wherein the trigger covers the trigger slot to reduce the chances of a user's finger being pinched within the trigger slot.

(26) In aspects according to the present disclosure, a switch assembly is disposed on one of the first or second shaft members and is configured to be engaged by the other of the first or second shaft members when the jaw members are approximated to move the switch assembly between a deactivated position and an activated position to control delivery of electrosurgical energy to the jaw members.

(27) In aspects according to the present disclosure, a knife return spring is operably coupled to the knife deployment mechanism and is configured to bias the knife toward the retracted position. In other aspects according to the present disclosure, the knife lockout includes a slot defined in the flange configured to operably engage a lock pin disposed in the knife deployment mechanism to prevent movement of the knife when engaged.

(28) In aspects according to the present disclosure, upon approximation of the first and second shaft members, the flange is configured to dislodge the slot from engagement with the lock pin of the knife deployment mechanism to allow selective actuation of the knife. In other aspects according to the present disclosure, the flange is connected to the first shaft member by a sleeve. In yet other aspects according to the present disclosure, the flange is connected to the first shaft member by a flange pin.

(29) In aspects according to the present disclosure, the flange includes an elongated shaft fixed at a distal end thereof to the first shaft member and, upon approximation of the first and second shaft members, the elongated shaft of the flange is configured to cantilever or flex to dislodge the lock pin from the slot defined therein. In other aspects according to the present disclosure, upon opening of the first and second shaft members relative to one another the bias of the elongated shaft reseats the lock pin within the slot. In still other aspects according to the present disclosure, the knife deployment mechanism includes an elongated slot defined therein to allow reciprocation of the lock pin therein.

(30) In aspects according to the present disclosure, the flange includes a ramp to facilitate reseating the lock pin within the slot of the flange upon return of the knife deployment mechanism.

(31) An electrosurgical forceps provided in accordance with additional aspects of the present disclosure includes first and second shaft members each having a jaw member disposed at a distal end thereof, the first and second shaft members configured to rotate about a pivot to move the jaw members between an open position and a closed position, the first and second shaft members defining a longitudinal axis therebetween. A knife deployment mechanism is disposed within the first shaft member and includes a trigger moveable along the longitudinal axis to deploy a knife operably coupled thereto between a retracted position relative to the jaw members and an extended position between the jaw members. A knife lockout is disposed within the first shaft member in oppositional alignment with the second shaft member and is configured to move upon approximation of the first and second shaft members between an engaged position preventing deployment of the knife and a disengaged position allowing deployment of the knife. A knife kickout mechanism is configured to force the knife forward upon movement of the first and second shaft members from an approximated position to a more open position, the knife kickout mechanism including a flange depending from the knife deployment mechanism in oppositional alignment with the second shaft member wherein, upon approximation of the first and second shaft members and actuation of the knife deployment mechanism in a first direction, the knife kickout rides within a slot defined within the second shaft member to abuttingly engage a ramp defined in the slot and wherein, upon opening of the first and second shaft members relative to one another, the ramp forces the knife kickout mechanism in an opposite direction to facilitate return of the knife deployment mechanism to an unactuated position.

(32) An electrosurgical forceps provided in accordance with additional aspects of the present disclosure includes first and second shaft members each having a jaw member disposed at a distal end thereof, the first and second shaft members configured to rotate about a pivot to move the jaw members between an open position and a closed position, the first and second shaft members defining a longitudinal axis therebetween. A knife deployment mechanism is disposed within the first shaft member and includes a trigger moveable along the longitudinal axis to deploy a knife operably coupled thereto between a retracted position relative to the jaw members and an extended position between the jaw members. A knife lockout is configured to move upon approximation of the first and second shaft members between an engaged position preventing deployment of the

knife and a disengaged position allowing deployment of the knife, the knife lockout including a flange operably connected to the first shaft member and depending therefrom in opposition to the second shaft member such that approximation of the first and second shaft members forces the flange against the second shaft member to disengage the knife lockout to allow actuation of the knife. A knife kickout mechanism is disposed within the first shaft member in oppositional alignment with the second shaft member and is configured to force the knife forward upon movement of the first and second shaft members from an approximated position to a more open position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various aspects and features of the present electrosurgical forceps are described hereinbelow with reference to the drawings wherein like numerals designate identical or corresponding elements in each of the several views;
- (2) FIG. 1A-1F are various views of a prior art electrosurgical forceps including a knife lockout mechanism;
- (3) FIG. 2A is a side view of a first iteration of an electrosurgical forceps according to the present disclosure shown superimposed upon the forceps of FIG. 1A illustrating the same rotating trigger mechanism as the trigger mechanism of FIG. 1A shown overhanging a bottom shaft of the presently disclosed forceps;
- (4) FIG. 2B is a side view of a second iteration of the electrosurgical forceps of FIG. 2A showing a linear trigger mechanism prior to actuation thereof;
- (5) FIG. 2C is a side view of the electrosurgical forceps of FIG. 2B after actuation of a trigger thereof;
- (6) FIG. 3A is a partial, front perspective view of the forceps of FIG. 2B showing the internal components of a link-driven knife deployment mechanism according to the present disclosure;
- (7) FIG. 3B is a side view of the forceps shown in FIG. 3A including a trigger carrier disposed atop the link-driven knife deployment mechanism according to the present disclosure;
- (8) FIG. 3C is a rear perspective view of another embodiment of a link-driven knife deployment mechanism according to the present disclosure;
- (9) FIG. 3D is a side view of another embodiment of a link-driven knife deployment mechanism according to the present disclosure;
- (10) FIG. 4A is a partial, front perspective view of the forceps of FIG. 2B showing the internal components of a rack and gear-driven knife deployment mechanism according to the present disclosure;
- (11) FIG. 4B is a rear perspective view of another embodiment of a gear-driven knife deployment mechanism according to the present disclosure;
- (12) FIG. 4C is a top perspective view of another embodiment of a gear-driven knife deployment mechanism according to the present disclosure;
- (13) FIG. 4D is a side view of another embodiment of a compound gear-driven knife deployment mechanism according to the present disclosure;
- (14) FIG. 5A is a side perspective view of the forceps and gear-driven knife deployment mechanisms of FIG. 4A showing a knife kickout according to the present disclosure;
- (15) FIG. 5B is a side perspective view of another embodiment of a knife kickout according to the present disclosure;
- (16) FIGS. 6A-6B are side views of the knife kickout of FIG. 5A illustrating operational movement thereof;
- (17) FIGS. 7A-7B are side views of another embodiment of a knife kickout according to the

present disclosure illustrating operational movement thereof;

(18) FIGS. **8A-8B** are side views of another embodiment of a knife kickout according to the present disclosure illustrating operational movement thereof;

(19) FIG. **9** is a side view of the knife lockout according to the present disclosure;

(20) FIG. **10** is a side view of another embodiment of a knife lockout according to the present disclosure; and

(21) FIG. **11** is a side view of another embodiment of a knife lockout according to the present disclosure.

DETAILED DESCRIPTION

(22) The present disclosure describes electrosurgical forceps for grasping, treating, and/or dividing tissue. The forceps includes two shafts each having a jaw member disposed at a distal end thereof and movable between open and closed positions to grasp tissue. The electrosurgical forceps also includes a knife configured to divide grasped tissue following treatment of the tissue (e.g., a tissue seal cycle). A knife lockout works in conjunction with the shafts to prevent deployment of the knife prior to the shafts reaching a sufficiently-approximated position corresponding to a sufficiently-closed position of jaw members as well as to prevent deployment of the knife during treatment of tissue.

(23) Referring generally to FIGS. **1A-1F**, a forceps **100** provided in accordance with the present disclosure includes first and second shafts **110**, **120** each having a proximal end portion **112a**, **122a** and a distal end portion **112b**, **122b**. An end effector assembly **200** of forceps **100** includes first and second jaw members **210**, **220** extending from distal end portions **112b**, **122b** of shafts **110**, **120**, respectively. Forceps **100** further includes a pivot member **130** pivotably coupling first and second shafts **110**, **120** with one another, a knife **140** (FIG. **1B**), a knife deployment mechanism **150** for selectively deploying knife **140** relative to end effector assembly **200**, a knife lockout **170** (FIGS. **1D-1F**) for preventing deployment of knife **140** prior to sufficient closure of jaw members **210**, **220**, and a switch assembly **180** including a depressible activation button **183** for enabling the selective supply of electrosurgical energy to end effector assembly **200**. An electrosurgical cable **300** electrically couples forceps **100** to a source of energy (not shown), e.g., an electrosurgical generator, to enable the supply of electrosurgical energy to jaw members **210**, **220** of end effector assembly **200** upon activation of switch assembly **180**.

(24) The internal working components of the prior art forceps of FIGS. **1A-1F** and, in particular, the inner-working components of the knife lockout **170** are disclosed in commonly-owned U.S. patent application Ser. No. 15/617,283, the entire contents of which being incorporated by reference herein.

(25) Continuing with reference to FIGS. **1A-1F**, knife deployment mechanism **150** is coupled to shaft **110** and generally includes a pair of opposed triggers **152** (FIG. **1D**) extending from either side of shaft **110**, a first linkage **154**, a second linkage **156**, and a biasing spring **158**. Knife deployment mechanism **150** is disposed within outer housing **116** of shaft **110** with the exception of opposed triggers **152** which extend from either side of outer housing **116**. First linkage **154** is configured for positioning on one side of inner frame **114** (FIG. **1C**) of shaft **110** and includes a pair of integral (or otherwise engaged) pivot bosses (not shown) extending from either side thereof at a first end portion of first linkage **154**. Each pivot boss enables engagement of opposed triggers **152** therewith on either side of shaft **110**, e.g., via press-fitting, adhesion, or other suitable engagement.

(26) A proximal end portion of second linkage **156** is pivotably coupled to first linkage **154** at a second end portion of first linkage **154**. A distal end portion of second linkage **156** is pivotably coupled to knife **140** via a pivot pin **163** (FIG. **1F**). Pivot pin **163** may be integrally formed with second linkage **156**, e.g., as a post extending therefrom, or may be a separate component from second linkage **156**. Pivot pin **163** extends transversely through a longitudinal slot **115e** (FIG. **1E**) of inner frame **114** of shaft **110** such that pivot pin **163** is constrained to longitudinal movement within longitudinal slot **115e**. Second linkage **156** is disposed on one side of inner frame **114**,

which may be the same side as first linkage **154** or the opposite side (as shown). In either configuration, pivot pin **163** extends from second linkage **156** and through longitudinal slot **115e** such that a portion of pivot pin **163** protrudes laterally from the opposite side of inner frame **114**. (27) Biasing spring **158** may be configured as an extension spring or other suitable biasing spring **158**. A distal end portion of biasing spring **158** is engaged to first linkage **154** and a proximal end portion of biasing spring **158** is engaged to a support plate **166** (FIG. 1C). Spring **158** may be engaged to any moveable components of the knife deployment mechanism **150**. Support plate **166** includes handle **118** of shaft **110** integrally formed therewith or otherwise engaged thereto, and may be secured within outer housing **116** in any suitable fashion, e.g., via protrusion-aperture engagement. Support plate **166** provides increased structural support to shaft **110** to inhibit splaying of shafts **110**, **120** during use. Shaft **120** similarly includes a support plate **168** integrally formed with or otherwise engaging handle **128** of shaft **120** and secured to outer housing **126**, although support plate **168** need not extend distally as with support plate **166**.

(28) Biasing spring **158** biases first linkage **154** towards a first orientation, corresponding to the un-actuated position of triggers **152** and the proximal-most position of second linkage **156**, thereby biasing knife **140** towards a retracted position (e.g., a proximal-most position of knife **140**). Upon rotation of either of triggers **152** relative to shaft **110**, first linkage **154** is rotated against the bias of biasing spring **158** to thereby urge second linkage **156** distally such that pivot pin **163** is driven distally through longitudinal slot **115e** (FIG. 1E) to urge knife **140** from the retracted position towards an extended position, wherein knife **140** extends through a slot defined in pivot member **130**, a channel of body plate **115**, and knife channels of jaw members **210**, **220**.

(29) In use, a distal portion of knife **140** is configured to reciprocate through the slot of pivot member **130** to translate through knife channels of jaw members **210**, **220** in response to actuation of either trigger **152**. Knife deployment mechanism **150** is operably positioned on shaft **110** and relative to shaft **120** such that triggers **152** only slightly extend beyond the height dimension of forceps **100** in the vicinity of triggers **152**, in the furthest-approximated position of shafts **110**, **120**. As a result of this configuration, forceps **100** benefits from a low-profile design that reduces the chances of triggers **152** catching on the surgeon, patient, or on nearby objections during use and/or as forceps **100** is inserted and withdrawn from the surgical site.

(30) Turning to FIGS. 1D-1F, knife lockout **170** works in conjunction with shafts **110**, **120** to prevent deployment of knife **140** prior to shafts **110**, **120** reaching a sufficiently-approximated position corresponding to a sufficiently-closed position of jaw members **210**, **220**. Knife lockout **170** includes a body **172** (FIGS. 1A-1C) that is disposed about a portion of the inner frame **114** of shaft **110** and forms a portion of outer housing **116** of shaft **110**. More specifically, as shown in FIG. 1A, body **172** of knife lockout **170** defines a complementarily-shaped abutting surface with the abutting surface of the adjacent other component(s) of housing **116** such that housing **116** defines a substantially continuous outer surface.

(31) Knife lockout **170** further includes a cantilever arm **174** extending proximally from body **172**. Cantilever arm **174** and body **172** may be integrally formed, e.g., via injection molding, or may be attached in any other suitable fashion. Cantilever arm **174** extends along inner frame **114** of shaft **110** on an opposite side of inner frame **114** as compared to second linkage **156** of knife deployment mechanism **150**. Cantilever arm **174** defines a relatively narrowed configuration to permit flexing of cantilever arm **174**. A finger **176** integrally formed with cantilever arm **174** extends generally perpendicularly from a free end of cantilever arm **174** and through an opening defined in outer housing **116** of shaft **110** towards shaft **120**. A first stop **179** is defined at the junction of cantilever arm **174** and finger **176**. First stop **179** protrudes from cantilever arm **174** and defines an angled distal wall **179a** and a vertical proximal wall **179b**. The finger **176** includes a second stop **175** extending distally from a vertical distal wall **173** of finger **176**. The second stop **175** defines a vertical proximal wall **177** that is generally parallel to vertical distal wall **173** of finger **176**. A nook **178** is defined between vertical proximal wall **177** of second stop **175** and vertical distal wall **173**

of finger **176**.

(32) Referring to FIG. **1D**, with shafts **110**, **120** sufficiently spaced-apart from one another and jaw members **210**, **220** in the open position, finger **176** of knife lockout **170** is spaced-apart from outer housing **126** of shaft **120** such that cantilever arm **174** is disposed in an at-rest position. In its at-rest position, cantilever arm **174** extends along and in a generally parallel orientation relative to longitudinal slot **115e** of inner frame **114** of shaft **110**. Further, vertical proximal wall **179b** of first stop **179** is disposed at the proximal end portion of longitudinal slot **115e** and prevents distal advancement of pivot pin through longitudinal slot **115e** in the at-rest position of cantilever arm **174** and, accordingly, prevents deployment of knife **140**.

(33) Referring to FIG. **1E**, in order to disengage knife lockout **170** to permit deployment of knife **140**, shafts **110**, **120** are sufficiently approximated such that jaw members **210**, **220** are moved to the closed position (e.g., to grasp tissue therebetween) and a portion of outer housing **126** of shaft **120** contacts finger **176** of knife lockout **170** to urge finger **176** further into housing **116** of shaft **110**. However, as shown in the configuration of FIG. **1E**, shaft **110** is sufficiently spaced from shaft **120** such that outer housing **116** of shaft **110** is spaced from or otherwise out of engagement with depressible button **183** of switch assembly **180** such that depressible button **183** is not depressed to activate switch assembly **180** for initiating the supply of energy from the energy source (not shown) to jaw members **210**, **220**. As finger **176** is urged further into housing **116** of shaft **110**, cantilever arm **174** is flexed such that vertical proximal wall **179b** of first stop **179** is removed from the distal path of pivot pin **163**. Once this has been achieved, knife deployment mechanism **150** may be actuated, as detailed above, to advance pivot pin **163** distally through slot to move knife **140** from the retracted position towards the extended position.

(34) Should shafts **110**, **120** be moved apart from one another sufficiently such that shaft **120** no longer urges finger **176** to flex cantilever arm **174**, cantilever arm **174** is resiliently returned to its at-rest position. If knife **140** is disposed in the retracted position at this point, vertical proximal wall **179b** is returned to block the distal path of pivot pin **163**. However, if knife **140** is disposed in the deployed position or a partially-deployed position, the return of cantilever arm **174** to its at-rest position does not block the distal path of pivot pin **163** via vertical proximal wall **179b**. Rather, upon subsequent return of knife **140** to the retracted position, pivot pin **163** is moved proximally and into contact with angled distal wall **179a** of first stop **179**, camming therealong and urging cantilever arm **174** to flex from the at-rest position sufficiently so as to enable pivot pin **163** to return to the proximal end of longitudinal slot **115e**.

(35) Once pivot pin **163** reaches this position, cantilever arm **174** is returned to the at-rest position and, as a result, vertical proximal wall **179b** is returned to blocking the distal path of pivot pin **163**, thereby resetting knife lockout **170** to prevent movement of knife **140** from the retracted position towards the extended position until shafts **110**, **120** are once again sufficiently approximated. The biasing force of biasing member **158** is sufficient to move pivot pin **163** proximally to deflect cantilever arm **174** and reset knife lockout **170** as detailed above. As such, resetting of knife lockout **170** occurs automatically (if shafts **110**, **120** are sufficiently spaced-apart) upon return of knife **140** to the retracted position.

(36) Referring to FIGS. **1E** and **1F** to activate switch assembly **180** to initiate the supply of energy from the energy source (not shown) to jaw members **210**, **220** for sealing tissue grasped between jaw members **210**, **220**, shafts **110**, **120** are further approximated from the approximated position illustrated in FIG. **1E** such that finger **176** is urged further into housing **116** of shaft **110** and depressible button **183b** is engaged and depressed by a portion of outer housing **116** of shaft **110** to activate switch assembly **180** (FIG. **1F**).

(37) As finger **176** is urged further into housing **116** of shaft **110**, cantilever arm **174** is further flexed such that vertical proximal wall **179b** of first stop **179** remains removed from the distal path of pivot pin **163** and second stop **175** is urged further into housing **116** of shaft **110** such that the portion of pivot pin **163** that extends from second linkage **156** through longitudinal slot **115e** is

received within nook **178** of second stop **175**. Once pivot pin **163** is received within nook **178**, vertical proximal wall **177** of second stop **175** prevents distal advancement of pivot pin **163** through longitudinal slot **115e** and, accordingly, prevents movement of knife **140** through jaw members **210**, **220** during activation of switch assembly **180**. In this manner, premature cutting of tissue during delivery of energy to tissue via jaw members **210**, **220** (e.g., prior to completion of a tissue sealing cycle) is prevented.

(38) Once a tissue sealing cycle is complete, switch assembly **180** may be deactivated by returning shafts **110**, **120** from an energy delivery position illustrated in FIG. **1F** to the approximated position illustrated in FIG. **1E** such that jaw members **210**, **220** remain in the closed position and depressible button **183b** is no longer depressed by outer housing **116** of shaft **110**. Upon returning to the approximated position illustrated in FIG. **1E**, cantilever arm **174** remains sufficiently flexed such that vertical proximal wall **179b** of first stop **179** is removed from the distal path of pivot pin **163**.

(39) Accordingly, knife deployment mechanism **150** may be actuated, as detailed above, to advance pivot pin **163** distally through slot **115e** to move knife **140** from the retracted position towards the extended position to cut tissue grasped between jaw members **210**, **220** (e.g., subsequent to completion of sealing the grasped tissue). Following cutting of the grasped tissue, shafts **110**, **120** may be moved apart from one another, as detailed above, to the spaced-apart position illustrated in FIG. **1D** such that cantilever arm **174** is resiliently returned to its at-rest position to reset knife lockout **170** to prevent movement of knife **140** from the retracted position towards the extended position.

(40) Details relating to the operation of the switch assembly **180** are disclosed in commonly-owned U.S. patent application Ser. No. 15/617,283, the entire contents of which being incorporated by reference herein.

(41) Cantilever arm **174** in use, functions as follows: when the shaft members **110**, **120** are disposed in an open position, e.g., the jaw members **210**, **220** are disposed in an open position, the blade **140** is prevented from being actuated as described above with respect to FIG. **1D**. Upon initial closure of the shaft members **110**, **120**, the blade **140** may be actuated as described above with reference to FIG. **1E**. Upon full actuation of the shaft members **110**, **120**, the blade is once again prevented from being actuated as described above with respect to FIG. **1F**.

(42) FIG. **2A** shows another embodiment of a forceps **1000** according to the present disclosure. Forceps **1000** is shown for illustrative purposes and includes opposing shafts **1110** and **1120** including an end effector assembly **1200** disposed at a distal end thereof. Shafts **1110** and **1120** are moveable via handles **1118**, **1128** about a pivot **1130** to open and close the end effector assembly **1200**. A trigger **1152** is disposed on shaft **1110** and is rotatable to deploy a knife (not shown) for cutting tissue much in the same fashion as described above with respect to forceps **100** of FIGS. **1A-1F**. Forceps **1000** is shown superimposed atop the frame of forceps **100** to illustrate how a distal end **1152a** of trigger **1152** projects beyond the periphery of shaft **1120** when disposed in an unactuated condition. As can be appreciated, designing a forceps in which the trigger projects from beyond the periphery of the shaft frame is not desirable.

(43) FIGS. **2B-2C** show a lower profile, linear trigger design which is configured to inhibit the trigger from catching on the surgeon, patient, or on nearby obstructions during use and/or as forceps **2000** is inserted and withdrawn from the surgical site. More particularly, FIGS. **2B-2C** show forceps **2000** having trigger **2150** that is actuated linearly along longitudinal axis "A-A" defined between shafts **2110**, **2120** to advance a knife (not shown, but see knife **140**) through tissue disposed between jaw members **2210**, **2220** of end effector assembly **2200**. As such, the trigger **2150** does not extend beyond the periphery of either shaft **2110**, **2120** during the range of linear motion.

(44) It is important to note that the various previously-described components are not described with reference to the remaining figures for the purposes of brevity and only those components necessary

for each figure are described, however, it is intended that the former components or variations thereof may be used interchangeably with the remaining figures.

(45) Referring back to FIGS. 2B-2C, FIG. 2B shows the linear trigger **2150** in an un-actuated position wherein the knife (not shown, but see knife **140**) is disposed in a retracted position within a knife channel (not shown) defined between the jaw members **2210**, **2220**. In this position, a trigger channel **2155** is exposed. FIG. 2C shows the trigger **2150** in a proximal, actuated position to deploy the knife between jaw members **2210**, **2220** to cut tissue. In this position, the trigger **2150** covers the trigger channel **2155** to reduce the chances of pinching a surgical glove or finger during repeated actuation. Trigger **2150** is symmetric on both sides of shaft **2110** allowing actuation by right or left-handed surgeons.

(46) With the reduced profile of the forceps **2000**, the internal working components of the knife deployment mechanism need to be slightly modified compared to the knife deployment mechanisms described above. Various types of deployment mechanism are envisioned and can generally be classified as linkage-type deployment mechanisms as shown in FIGS. 3A-3D and gear-like deployment mechanisms as shown in FIGS. 4A-4D.

(47) Referring initially to FIGS. 3A-3B, a linkage-type knife deployment mechanism **3150** is shown and includes a knife carrier **3170**, a series of linkages **3160**, **3180**, **3190** and a knife return spring **3115** that cooperate to smoothly advance the knife (not shown) in a linear motion upon actuation of trigger **2150** (FIGS. 2B-2C). More particularly, trigger **2150** connects to link **3190** via pin **3195** which, in turn, operably connects to crank link **3180** via pivot **3185**. Pivot **3185** is configured to slide linearly in slot **3140** defined in knife support **3114**. Crank link **3180** is operably connected to link **3160** via pivot **3175** which is supported within arcuate channel **3112** defined within shaft **3110**. Link **3160** is operably connected to knife carrier **3170** via pin **3165**. Pin **3165** is configured to translate knife carrier **3170** within slot **3120** defined in a distal end of knife support **3114**. Knife support **3114** is, in turn, operably connected to knife (not shown, but see knife **140**).

(48) One end **3115a** of knife return spring **3115** connects to crank link **3180** proximate link **3160** and moves concurrently therewith upon actuation of trigger **2150**. The other end **3115b** of spring **3115** is secured to shaft **3110**.

(49) The linkage design with a linear trigger **2150** allows the forceps **2000** height to decrease, requiring less housing constraints while maintaining a similar mechanical advantage to previous designs.

(50) In use, when trigger **2150** is actuated (pulled back proximally), pin **3195** moves link **3190** proximally which, in turn, moves pivot **3185** proximally within slot **3140**. As pivot **3185** moves proximally within slot **3140**, a distal end **3180a** of crank link **3180** moves therewith causing pin **3183** to move proximally and rotate clockwise within slot **3130** which, in turn, forces the proximal end **3180b** of crank link **3180** distally. Movement of proximal end **3180b** of crank link **3180** distally rotates link **3160** distally within arcuate slot **3112** against the bias of return spring **3115**. Distal movement of link **3160** forces knife carrier **3170** distally within slot **3120** to advance knife (not shown) through tissue.

(51) Once actuated, the force of spring **3115** reverses the motion of the knife carrier **3170** and links **3160**, **3180**, **3190** to return the trigger **2150** distally back to an unactuated position. As explained in more detail below, if the knife (not shown) is stuck in the knife channel between jaw members **2210**, **2120** or on tissue, a knife kickout may be used to force the knife proximally as the shafts **2110**, **2120** are opened (See FIGS. 6A-8B).

(52) FIG. 3B shows another embodiment of the knife deployment mechanism **3150** for use with forceps **2000** which includes a trigger carrier **3300** that operably connects to the trigger **2150**. Trigger carrier **3300** includes a cam slot **3310** defined therein that resides in general perpendicular registration with slot **3140** in knife carrier **3114** and that is configured to slidably receive pivot **3185** therein. Upon actuation of the trigger **2150**, trigger carrier **3300** is moved proximally and the pivot **3185** rides along the cam slot **3310** in a general perpendicular direction. Utilizing the cam slot

3310 allows for smoother and more consistent actuation of trigger **2150**.

(53) FIG. 3C shows another embodiment of the knife deployment mechanism **3150** for use with forceps **2000** which includes a guide slot **3163** defined within a proximal end of link **3160** that allows pivot **3175** to ride therein during actuation of trigger **2150**. Upon actuation of the trigger **2150**, link **3170** is moved proximally to pivot link crank **3180** distally which, in turn, forces link **3160** distally to advance knife (not shown). Utilizing the guide slot **3163** allows for smoother and more consistent actuation of the trigger **2150** and knife through their range of motion.

(54) FIG. 3D shows another embodiment of the knife deployment mechanism **3150** for use with forceps **2000** which includes a knife lockout mechanism **3370** (shown in phantom) for use with the knife deployment mechanism **3150**. Knife lockout mechanism **3370** is similar to knife lockout **170** of FIGS. 1D-1F and, as such, is only described in brief detail herein. Knife lockout mechanism **3370** includes a flange **3355** disposed in operative engagement with the knife carrier **3170** and depending therefrom. When the trigger **2150** is in an unactuated position, flange **3355** is configured to project downwardly relative thereto in alignment with shaft **2120** of forceps **2000**. Upon closing of the handles **2118**, **2128**, the flange **3355** abuts against shaft **2120** and is forced inwardly toward shaft **2110** to disengage the flange **3355** from the knife carrier **3170**. Once disengaged, the trigger **2150** is free to actuate the knife (not shown). When the trigger **2150** is returned to an unactuated position and the handles **2118**, **2128** are moved away from one another, the flange **3355** re-engages the knife carrier **3170** to prevent translation of the knife.

(55) Referring to FIGS. 4A and 4B, a gear-type knife deployment mechanism **4150** is shown and includes a blade rack **4170**, a trigger rack **4160** and a gear **4155** disposed therebetween. Gear **4155** is configured to reverse direction between the two racks **4160**, **4170** and may also be utilized to amplify overall force or distance therebetween. Blade rack **4170** is operably coupled to the trigger **2150** (FIG. 2B) via a trigger carrier **4300**, which, in turn, includes one or more pins **4195** or other mechanical interfaces that engage the trigger **2150** directly such that movement of the trigger **2150** correspondingly moves the blade rack **4170**.

(56) Gear **4155** is mounted about pin **4157** to shaft **4110** and between racks **4160**, **4170**. Movement of one rack, e.g., rack **4170** causes the other rack, e.g., rack **4160**, to move in the opposite direction. Rack **4160** is operably couple to the knife carrier **4200**. As such, proximal movement of the trigger **2150** is converted to distal movement of the knife (not shown).

(57) FIG. 4B shows an alternative setup of a gear-type knife deployment mechanism **4150** showing slightly modified blade and trigger racks **4170**, **4160** for actuating the knife and return spring **4115** for facilitating the return of the knife. Return spring **4115** is operably coupled to the blade rack **4170** but may be coupled to either rack **4160**, **4170** depending upon a particular purpose.

(58) FIG. 4C shows yet another embodiment of a gear-type knife deployment mechanism **4150'** showing slightly modified blade and trigger racks **4170'**, **4160'** and two gears for actuating the knife (not shown). More particularly, a single gear **4155a** is operably associated with trigger rack **4160'** and a compound gear **4155b** is operably associated with blade rack **4170'**. Compound gear **4155b** allows the two gears **4155a**, **4155b** to align in general vertical registry while still providing the same reversing effect between the trigger **2150** and knife deployment. As can be appreciated, this may save valuable real estate within the shaft **4110** for additional components or a reduced profile.

(59) FIG. 4D shows another embodiment of a combination gear and lever-type knife deployment mechanism **5150** having a trigger rack **5160** operably coupled to a compound gear **5155c**, multiple single gears **5155a**, **5155b**, a knife link **5190**, and a knife carrier **5200**. As can be appreciated, this combination deployment mechanism **5150** can be configured to provide mechanical advantages for amplifying overall force applied to the knife or distance the knife travels with trigger **2150** actuation.

(60) FIG. 5A shows a knife kickout mechanism **4370** for use with the knife deployment mechanism **4150** of FIG. 4A. Knife kickout mechanism **4370** includes a flange **4355** disposed in operative

engagement with the knife carrier **4200** (See FIG. 4A) and depending therefrom. When the trigger **2150** is in an unactuated position, flange **4355** is configured to project downwardly relative thereto in alignment with shaft **2120** of forceps **2000**. Upon closing of the handles **2118**, **2128**, the flange **4355** is received within a slot defined in the shaft **2110** and is forced inwardly toward a kickout ramp **2121** (See FIG. 6A). When the trigger **2150** is returned to an unactuated position and the handles **2118**, **2128** are moved away from one another, the flange **4355** engages the kickout ramp **212** to urge the knife carrier **4200** forward facilitating knife carrier **4200** return.

(61) FIG. 5B shows a knife lockout mechanism **6370** for use with the knife deployment mechanism similar to those shown in FIGS. 3A-3D. Knife lockout mechanism **6370** is similar to knife lockout **170** of FIGS. 1D-1F and as such, is only described in brief detail herein. Knife lockout mechanism **6370** includes a flange **6355** disposed in rotational engagement with the knife carrier **6200**. When the trigger **2150** is in an unactuated position, flange **6355** is configured to project downwardly relative thereto in alignment with shaft **2120** of forceps **2000**. Upon closing of the handles **2118**, **2128**, the flange **6355** abuts against shaft **2120** and is forced inwardly toward shaft **2110** (See FIGS. 2B and 2C) and a distal end **6356** of the flange **6355** rotates out of engagement with a proximal hook portion **6201** of the knife carrier **6200**. Once disengaged, the trigger **2150** is free to actuate the knife (not shown). When the trigger **2150** is returned to an unactuated position and the handles **2118**, **2128** are moved away from one another, the distal end **6356** of the flange **6355** re-engages the hook portion **6201** of the knife carrier **6200** to prevent translation of the knife.

(62) Referring to FIGS. 6A-8B, various embodiments of a knife kickout are envisioned. The described knife kickout mechanisms are configured to work with many of the aforescribed forceps and internal components thereof, and as such, only those components necessary for an accurate understanding of the kickout are described in detail herein.

(63) FIGS. 6A-6B show one embodiment of a knife kickout **6310** for use with trigger carrier **6300**. Knife kickout **6310** is configured to depend from trigger carrier **6300** and align in vertical registration with an elongated kickout slot **2123** defined within shaft **2120**. Shaft **2120** also includes a kickout ramp **2121** defined at a proximal-most portion of the kickout slot **2123**. Upon approximation of handles **2118**, **2128** of forceps **2000** (FIGS. 2A-2C), shafts **2110**, **2120** are urged into close abutment with one another to either allow actuation of the knife (not shown) via trigger **2150** and/or activation of electrical energy via switch **2183**.

(64) As the trigger carrier **6300** is actuated (proximally), the kickout **6310** rides within slot **2123** of shaft **2120** into abutment with kickout ramp **2121**. Typically, upon release of the trigger **2150**, trigger carrier **6300** is supposed to automatically return to a distal-most position under the bias of knife return spring **6315**. If the knife (not shown) gets caught in the knife channel disposed between jaw members **2210**, **2220** or gets caught on tissue, the bias of the knife return spring **6315** may not be enough and the knife may remain in a deployed position.

(65) In this instance, and in order to kick out the knife and get it moving proximally out of the knife channels, the user simply begins to open the forceps **2000** causing the shafts **2110**, **2120** to move relative to one another and causing the kickout ramp **2121** to essentially “kick” the knife kickout **6310** forward as the handles **2118**, **2128** open. Forward motion of the knife kickout **6310**, in turn, forces the trigger carrier **6300** forward and, thus, forces the knife proximally.

(66) FIGS. 7A-7B show another embodiment of a knife kickout **7310** for use with trigger carrier **7300**. Knife kickout **7310** is configured to depend from trigger carrier **7300** and align in vertical registration with an elongated kickout slot **2123** defined within shaft **2120**. Shaft **2120** also includes a kickout ramp **2121** defined at a proximal-most portion of the kickout slot **2123**. Upon approximation of handles **2118**, **2128** of forceps **2000** (FIGS. 2A-2C), shafts **2110**, **2120** are urged into close abutment with one another to either allow actuation of the knife (not shown) via trigger **2150** and/or activation of electrical energy via switch **2183**.

(67) As the trigger carrier **7300** is actuated (proximally), the kickout **7310** rides within slot **2123** of shaft **2120** into abutment with kickout ramp **2121**. Typically, upon release of the trigger **2150**,

trigger carrier **7300** is supposed to automatically return to a distal-most position under the bias of knife return spring **7315**. If the knife (not shown) gets caught in the knife channel disposed between jaw members **2210**, **2220** or gets caught on tissue, the bias of the knife return spring **7315** may not be enough and the knife may remain in a deployed position.

(68) In this instance, and in order to kick out the knife and get it moving proximally out of the knife channels, the user simply begins to open the forceps **2000** causing the shafts **2110**, **2120** to move relative to one another and causing the kickout ramp **2121** to essentially “kick” the knife kickout **7310** forward as the handles **2118**, **2128** open. Forward motion of the knife kickout **7310**, in turn, forces the trigger carrier **7300** forward and, thus, forces the knife proximally.

(69) FIGS. **8A-8B** show another embodiment of a knife kickout **8310** for use with kickout link **8300**. Knife kickout **8310** is configured to depend from kickout link **8300** and align in vertical registration with an elongated kickout slot **2123** defined within shaft **2120**. Shaft **2120** also includes a kickout ramp **2121** defined at a proximal-most portion of the kickout slot **2123**. Upon approximation of handles **2118**, **2128** of forceps **2000** (FIGS. **2A-2C**), shafts **2110**, **2120** are urged into close abutment with one another to either allow actuation of the knife (not shown) via trigger **2150** and/or activation of electrical energy via switch **2183**. Kickout link **8300** is rotatably engaged at one end to link **8180** and positioned proximally of pivot **8195** a distal end of link **8190** such that movement of link **8190** pulls kickout link **8300** proximally along therewith.

(70) As mentioned above with respect to FIG. **3A**, as the trigger **2150** is actuated, kickout link **8300** is actuated (proximally), the kickout **8310** rides within slot **2123** of shaft **2120** into abutment with kickout ramp **2121**. Typically, upon release of the trigger **2150**, kickout link **8300** is supposed to automatically return to a distal-most position under the bias of knife return spring **8315**. If the knife (not shown) gets caught in the knife channel disposed between jaw members **2210**, **2220** or gets caught on tissue, the bias of the knife return spring **8315** may not be enough and the knife may remain in a deployed position.

(71) In this instance, and in order to kick out the knife and get it moving proximally out of the knife channels, the user simply begins to open the forceps **2000** causing the shafts **2110**, **2120** to move relative to one another and causing the kickout ramp **2121** to essentially “kick” the knife kickout **8310** and link **8180** forward as the handles **2118**, **2128** open. Forward motion of the knife kickout **8310**, in turn, forces the kickout link **8300** and link **8180** forward and, thus, forces the knife proximally.

(72) Referring to FIGS. **9-11**, various embodiments of a knife lockout are envisioned. The described knife lockout mechanisms are configured to work with many of the aforescribed forceps and internal components thereof, and as such, only those components necessary for an accurate understanding of the kickout are described in detail herein.

(73) FIG. **9** shows one embodiment of a knife lockout **9000** for use with, for example, forceps **2000** and knife deployment mechanism **4150** described in FIG. **4A**. Knife lockout **9000** is L-shaped and includes an elongated shaft **9050** connected at a distal end thereof to shaft **2110** by a flange pin **9175** and terminating at an opposite end thereof with a flange **9100** that depends in oppositional registry with shaft **2120**. The distal end **9105** of flange **9100** is configured to abut shaft **2120** upon approximation of shaft members **2110**, **2120** to force flange inwardly towards shaft **2110**.

(74) Knife deployment mechanism **4150** includes a locking pin **4365** operably engaged with trigger rack **4160** (see FIG. **4B**) that is configured to ride within an elongated slot **4375** defined therein upon actuation of the knife deployment mechanism **4150**. Flange **9100** includes a slot **9110** defined therein configured to seat locking pin **4365** therein to prevent actuation of the knife deployment mechanism **4150** when the shaft members **2110**, **2120** are disposed in an open position.

(75) In use, when the shaft members **2110**, **2120** are disposed in an open position relative to one another, the locking pin **4365** is seated within slot **9110** preventing movement of the knife deployment mechanism **4150**. When the shaft members **2110**, **2120** are approximated, the distal end **9105** of flange **9100** abuts shaft **2120** forcing the flange **9100** towards shaft **2110** and causing

the elongated shaft **9050** to flex about flange pin **9175**. As a result, locking pin **4365** is unseated or disengaged from slot **9110** allowing actuation of the knife deployment mechanism **4150**. Locking pin **4365** rides along elongated slot **4375** during actuation. Flange pin **9175** is anchored to shaft **2110** to bias flange **9100**. A conventional spring, e.g., torsion spring (not shown), may also be utilized for this purpose.

(76) After actuation of the knife deployment mechanism **4150**, the knife deployment mechanism **4150** and the trigger **2150** are released and returned under the bias of the knife return spring **4115** (See FIG. 4A). As a result thereof, locking pin **4365** returns along elongated slot **4375** to its unactuated position. Upon return, the locking pin **4365** engages ramp a **9115** disposed distally of slot **9110** which forces flange **9100** inwardly relative to shaft **2110** to allow the locking pin **4365** to reseat within slot **9110** re-locking the knife deployment mechanism **4150** and preventing movement thereof. Any type of conventional spring (not shown) may also be utilized for this purpose.

(77) FIG. 10 shows another embodiment of a knife lockout **10000** for use with, for example, forceps **2000** and knife deployment mechanism **3150** described in FIG. 3A. Knife lockout **10000** is L-shaped and includes an elongated shaft **10050** connected at a distal end thereof to shaft **2110** by a sleeve **10175** and terminating at an opposite end thereof with a flange **10100** that depends in oppositional registry with shaft **2120**. The distal end **10105** of flange **10100** is configured to abut shaft **2120** upon approximation of shaft members **2110**, **2120** to force flange inwardly towards shaft **2110**.

(78) Knife deployment mechanism **3150** includes a locking pin **3165** that operably engages link **3160** and knife carrier **3170** and that is configured to ride within an elongated slot **3375** defined in knife carrier **3170** upon actuation of the knife deployment mechanism **3150**. Flange **10100** includes a slot **10110** defined therein configured to seat locking pin **3165** therein to prevent actuation of the knife deployment mechanism **3150** when the shaft members **2110**, **2120** are disposed in an open position.

(79) In use, when the shaft members **2110**, **2120** are disposed in an open position relative to one another, the locking pin **3165** is seated within slot **10110** preventing movement of the knife deployment mechanism **3150**. When the shaft members **2110**, **2120** are approximated, the distal end **10105** of flange **10100** abuts shaft **2120** forcing the flange **10100** towards shaft **2110** and causing the elongated shaft **10050** to flex about sleeve **10175**. As a result, locking pin **3165** is unseated or disengaged from slot **10110** allowing actuation of the knife deployment mechanism **3150**. Locking pin **3165** rides along elongated slot **3375** during actuation.

(80) After actuation of the knife deployment mechanism **3150**, the knife deployment mechanism **3150** and the trigger **2150** are released and returned under the bias of the knife return spring **3115** (See FIG. 3A). As a result thereof, locking pin **3165** returns along elongated slot **3375** to its unactuated position. Upon return, the locking pin **3165** engages ramp a **10115** disposed distally of slot **10110** which forces flange **10100** inwardly relative to shaft **2110** to allow the locking pin **3165** to reseat within slot **10110** re-locking the knife deployment mechanism **3150** and preventing movement thereof.

(81) FIG. 11 shows another embodiment of a knife lockout **11000** for use with, for example, forceps **2000** and knife deployment mechanism **4150** described in FIG. 4A. Knife lockout **11000** is L-shaped and includes an elongated shaft **11050** connected at a proximal end thereof to shaft **2110** by a flange pin **11175** and terminating at an opposite end thereof with a flange **11100** that depends in oppositional registry with shaft **2120**. The distal end **11105** of flange **11100** is configured to abut shaft **2120** upon approximation of shaft members **2110**, **2120** to force flange inwardly towards shaft **2110**.

(82) Flange **11100** includes a boss **11110** disposed thereon therein configured to seat within a gear slot **4171** defined in gear rack **4170** to prevent actuation of the knife deployment mechanism **4150** when the shaft members **2110**, **2120** are disposed in an open position.

(83) In use, when the shaft members **2110**, **2120** are disposed in an open position relative to one

another, the boss **1110** is seated within gear slot **4171** preventing movement of the knife deployment mechanism **4150**. When the shaft members **2110**, **2120** are approximated, the distal end **11105** of flange **11100** abuts shaft **2120** forcing the flange **11100** towards shaft **2110** and causing the elongated shaft **11050** to rotate about flange pin **11175**. As a result, boss **11110** is unseated or disengaged from gear slot **4171** allowing actuation of the knife deployment mechanism **4150**. Boss **11110** remains out of the way during movement of the opposing racks **4170**, **4160** (FIG. 4A).

(84) After actuation of the knife deployment mechanism **4150**, the knife deployment mechanism **4150** and the trigger **2150** are released and returned under the bias of the knife return spring **4115** (See FIG. 4A). As a result thereof, boss **11110** re-engages gear slot **4171** re-locking the knife deployment mechanism **4150** and preventing movement thereof.

(85) The various embodiments disclosed herein may also be configured to work with robotic surgical systems and what is commonly referred to as “Telesurgery.” Such systems employ various robotic elements to assist the surgeon and allow remote operation (or partial remote operation) of surgical instrumentation. Various robotic arms, gears, cams, pulleys, electric and mechanical motors, etc. may be employed for this purpose and may be designed with a robotic surgical system to assist the surgeon during the course of an operation or treatment. Such robotic systems may include remotely steerable systems, automatically flexible surgical systems, remotely flexible surgical systems, remotely articulating surgical systems, wireless surgical systems, modular or selectively configurable remotely operated surgical systems, etc.

(86) The robotic surgical systems may be employed with one or more consoles that are next to the operating theater or located in a remote location. In this instance, one team of surgeons or nurses may prep the patient for surgery and configure the robotic surgical system with one or more of the instruments disclosed herein while another surgeon (or group of surgeons) remotely controls the instruments via the robotic surgical system. As can be appreciated, a highly skilled surgeon may perform multiple operations in multiple locations without leaving his/her remote console which can be both economically advantageous and a benefit to the patient or a series of patients.

(87) The robotic arms of the surgical system are typically coupled to a pair of master handles by a controller. The handles can be moved by the surgeon to produce a corresponding movement of the working ends of any type of surgical instrument (e.g., end effectors, graspers, knives, scissors, etc.) which may complement the use of one or more of the embodiments described herein. The movement of the master handles may be scaled so that the working ends have a corresponding movement that is different, smaller or larger, than the movement performed by the operating hands of the surgeon. The scale factor or gearing ratio may be adjustable so that the operator can control the resolution of the working ends of the surgical instrument(s).

(88) The master handles may include various sensors to provide feedback to the surgeon relating to various tissue parameters or conditions, e.g., tissue resistance due to manipulation, cutting or otherwise treating, pressure by the instrument onto the tissue, tissue temperature, tissue impedance, etc. As can be appreciated, such sensors provide the surgeon with enhanced tactile feedback simulating actual operating conditions. The master handles may also include a variety of different actuators for delicate tissue manipulation or treatment further enhancing the surgeon's ability to mimic actual operating conditions.

(89) From the foregoing and with reference to the various figure drawings, those skilled in the art will appreciate that certain modifications can also be made to the present disclosure without departing from the scope of the same. While several embodiments of the disclosure have been shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Therefore, the above description should not be construed as limiting, but merely as exemplifications of particular embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

Claims

1. An electrosurgical forceps, comprising: first and second shaft members each having a jaw member disposed at a distal end thereof, the first and second shaft members configured to rotate about a pivot to move the jaw members between an open position and a closed position, the first and second shaft members defining a longitudinal axis therebetween; a knife deployment mechanism disposed within the first shaft member and including a trigger moveable along the longitudinal axis to deploy a knife operably coupled thereto between a retracted position relative to the jaw members and an extended position between the jaw members, the knife deployment mechanism including first and second rack members operably coupled to one another by a gear disposed therebetween, the trigger operably connected to the first rack member and the knife operably coupled to the second rack member such that movement of the trigger moves the knife in an opposite direction relative thereto; and a knife lockout configured to move upon approximation of the first and second shaft members between an engaged position preventing deployment of the knife and a disengaged position allowing deployment of the knife, the knife lockout including a flange and an elongated shaft that is pivotably connected to the first shaft member by a flange pin such that approximation of the first and second shaft members forces the second shaft member into the flange to pivot the flange and the elongated shaft relative to the first shaft member to disengage the knife lockout to allow actuation of the knife, wherein the knife lockout includes a slot defined in the flange configured to operably capture a lock pin disposed in the knife deployment mechanism to prevent movement of the knife when engaged.
2. The electrosurgical forceps according to claim 1, wherein the first shaft member includes a trigger slot defined therein, the trigger is configured to travel between a distal-most position wherein the trigger slot is exposed and a more proximal position wherein the trigger covers the trigger slot to reduce the chances of a user's finger being pinched within the trigger slot.
3. The electrosurgical forceps according to claim 1, further comprising a switch assembly disposed on one of the first or second shaft members and configured to be engaged by the other of the first or second shaft members when the jaw members are approximated to move the switch assembly between a deactivated position and an activated position to control delivery of electrosurgical energy to the jaw members.
4. The electrosurgical forceps according to claim 1, further comprising a knife return spring operably coupled to the knife deployment mechanism and configured to bias the knife toward the retracted position.
5. The electrosurgical forceps according to claim 4, wherein the knife return spring is operably coupled to at least one of the first or second rack members.
6. The electrosurgical forceps according to claim 1, wherein upon approximation of the first and second shaft members, the flange is configured to dislodge the lock pin from within the slot of the knife deployment mechanism to allow selective actuation of the knife.
7. The electrosurgical forceps according to claim 1, wherein upon approximation of the first and second shaft members, the elongated shaft is configured to cantilever or flex to pivot about the flange pin to dislodge the lock pin from within the slot.
8. The electrosurgical forceps according to claim 1, wherein upon opening of the first and second shaft members relative to one another a bias of the flange reseats the lock pin for capture within the slot.
9. The electrosurgical forceps according to claim 1, wherein the knife deployment mechanism includes an elongated slot defined therein to allow reciprocation of the lock pin therein.
10. The electrosurgical forceps according to claim 1, wherein the flange includes a ramp to facilitate reseating the lock pin for capture within the slot upon return of the knife deployment mechanism.

11. The electrosurgical forceps according to claim 1, further comprising a knife kickout mechanism configured to force the knife forward upon movement of the first and second shaft members from an approximated position to a more open position.

12. The electrosurgical forceps according to claim 11, wherein the knife kickout mechanism includes a flange depending from the knife deployment mechanism in oppositional alignment with the second shaft member wherein, upon approximation of the first and second shaft members and actuation of the knife deployment mechanism in a first direction, the knife kickout rides within a slot defined within the second shaft member to abuttingly engage a ramp defined in the slot and wherein, upon opening of the first and second shaft members relative to one another, the ramp forces the knife kickout mechanism in an opposite direction to facilitate return of the knife deployment mechanism to an unactuated position.
